

# Main Report — HW03 GUI and Usability Testing

Student ID: 23127334 Name: Nguyễn Thành Dân Exercise ID: HW02 Date: 2026-06-28

## 1. Scope and environment

The SUT is the local `eshop-sut` repository. The web frontend runs at `http://localhost:5173` and the API at `http://localhost:3000`.

| Scope                          | Route                 | Requirements       |
|--------------------------------|-----------------------|--------------------|
| Login                          | <code>/login</code>   | FR-02, FR-21–FR-24 |
| Order History (inside Profile) | <code>/profile</code> | FR-11, FR-21–FR-24 |

The selected usability flow is: sign in with a supplied account, independently locate the order history, then identify the latest order's ID, date, total, and current status. The student confirmed that this scope does not duplicate another group member's primary screen or flow.

## 2. Interface analysis

### User goals

- Sign in safely and understand whether authentication succeeded or failed.
- Recover from a typing or credential error without exposing account details.
- Locate personal order history after authentication.
- Identify the newest order and understand its date, total, and state.

### Main UI components and forms

Login contains an email field, password field, forgot-password link, submit button, registration link, and error feedback. Order History shares the Profile page with the profile-edit form and renders either an empty message or a table containing order ID, date, total, status, and an action.

### Navigation paths

`Header → Đăng nhập → /login → successful login → Home → profile link → /profile → Lịch sử đơn hàng`. Direct access and expired-session behavior must also be checked.

## Feedback and state changes

Important states are initial, invalid native form input, wrong credentials, temporary lock, successful login, order loading, empty history, populated history, API failure, and session expiration. Results must be observed at runtime; source inspection is used only to design checks.

## Risks

- Accessibility: semantic labels, focus order, error announcement, table structure, color-independent status, keyboard access, zoom and text scaling.
- Responsive design: the five-column order table may overflow narrow screens.
- Language: mixed Vietnamese and English labels may reduce clarity.
- Error recovery: authentication feedback must be timely, non-revealing, and positioned before submit according to FR-22.
- State ambiguity: an empty array may represent either no orders or an API failure unless distinct states are rendered.
- Localization: dates and VND amounts must use deterministic Vietnamese format.
- Theme and direction: dark mode and RTL are exploratory risks where supported.

## 3. GUI checklist design

---

The reviewed checklist contains 45 atomic GUI-focused items across both screens and all four interface aspects. Initial AI items are separated from human-added and refined items through separate versions. Security, backend authorization, and order business-rule checks were removed because they are outside this GUI assignment. Execution columns intentionally remain blank until the SUT is run.

The two-version audit trail is:

- `gui-checklist-v0-agent.xlsx` : the unreviewed agent-generated baseline.
- `gui-checklist-v1-reviewed.xlsx` : 45 GUI items after standards-based human review.

Neither submitted workbook contains a `Source` column. The differences and reasons for human additions are documented separately in `human-review.md`.

## Critical review of AI-generated items

The detailed V0-to-V1 comparison, explanation for every added item, scope migration, and lessons learned are documented in `task-1-gui-checklist/human-review.md`.

## 4. Checklist execution summary

---

The 45-item V1 checklist was executed with Playwright 1.55.0 on Chromium against the running local frontend and backend. Test setup created five orders through authenticated APIs so all five visible

order-status labels could be assessed.

| Result  | Count |
|---------|-------|
| Passed  | 15    |
| Failed  | 28    |
| Blocked | 2     |
| Total   | 45    |

Failed items contain an actual result, note, screenshot, and consolidated Bug ID. The two Blocked items require a real screen reader to verify reading order or live status announcement; no result was inferred from source code.

## 5. GUI bugs

---

The 28 Failed items were consolidated by root cause into 12 GUI bugs. Major findings include exposed password text, inaccessible form semantics, missing cancel confirmation, absent loading/error states, and page-wide overflow of the order table. See [task-1-gui-checklist/bug-report.md](#).

All 12 bugs were reported on GitHub as issues #18–#29 with the [homework3](#) label. Links are recorded in the checklist and [github-issues-links.md](#).

## 6. Usability plan

---

The moderated evaluation uses SUS with seven real participants outside HW03. The participant receives a goal rather than UI instructions. A pilot must be completed and reviewed before official sessions begin.

## 7. Pilot and seven-session results

---

The pilot was completed as a moderator dry-run before the official sessions and marked [READY](#). Seven real participants (P01–P07) recruited outside HW03 each completed the flow in a moderated session; every session was screen-recorded with consent (links in [task-2-usability/recording-links.md](#)).

| Metric                | Value                         |
|-----------------------|-------------------------------|
| Completion            | 7/7 (100%)                    |
| Unassisted completion | 5/7 (P01, P03, P04, P05, P07) |
| Assisted completion   | 2/7 (P02, P06)                |
| Mean completion time  | 287.3 s (4:47)                |
| Mean SUS              | 66.4/100                      |

| Metric                                                   | Value          |
|----------------------------------------------------------|----------------|
| Total errors / wrong turns / hesitations / interventions | 1 / 4 / 19 / 2 |

Per-participant raw SUS answers and scores are in [task-2-usability/sus-summary.csv](#) ; observation notes, probe answers, and evidence links are under [task-2-usability/sessions/](#) .

## 8. Usability analysis

All seven participants eventually reported the four FR-11 fields, but 6/7 struggled to discover where order history lived — the account greeting was not recognised as a link — making discoverability the dominant, critical finding. The severity-ranked findings ( [task-2-usability/severity-ranked-findings.md](#) ):

| Rank | ID      | Severity | Summary                                                      |
|------|---------|----------|--------------------------------------------------------------|
| 1    | USAB-01 | Critical | Order-history entry point not discoverable (7/7)             |
| 2    | USAB-07 | Major    | Order history lacks product detail / actions (5/7)           |
| 3    | USAB-04 | Major    | Password not masked + EN/VI mix on login (4/7)               |
| 4    | USAB-03 | Major    | Ambiguous US date format, no time shown (4/7)                |
| 5    | USAB-06 | Major    | Table sort order not labelled (3/7)                          |
| 6    | USAB-05 | Minor    | Unexpected keyboard focus order (1/7)                        |
| 7    | USAB-08 | Minor    | Cosmetic: red total colour, prominent exit, order code (2/7) |

Mean SUS 66.4 sits just below the ~68 acceptability benchmark, consistent with the discoverability friction. The primary recommendation is a clearly labelled "Đơn hàng của tôi" navigation entry, plus deterministic Vietnamese date/currency formatting and a masked password field.

## 9. Cross-platform results

The three required platforms were tested on **BrowserStack Live**, running the Task-2 flow **Login** → **Hồ sơ** → **Lịch sử đơn hàng (FR-11)** against the local SUT exposed through Cloudflare tunnels (frontend + backend). Seven screenshots were captured; each shows the OS / browser / device in the BrowserStack toolbar, the SUT tunnel URL, and the student identity [23127334@hcmus.edu.vn](#) (typed into Username on login; shown as the Profile email plus the greeting "Chào, Nguyễn Thành Dân" and "Họ Tên" field on the Order-History screens).

| Platform                | OS / device                                | Screens                    | Screenshots                                                                                   | Result                             |
|-------------------------|--------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------|------------------------------------|
| Google Chrome 147       | macOS desktop                              | Login;<br>Order<br>History | <a href="#">chrome1.png</a> ;<br><a href="#">chrome2.png</a>                                  | Rendered correctly                 |
| Mozilla Firefox 144–145 | Windows 11 desktop                         | Login;<br>Order<br>History | <a href="#">firefox1.png</a> ;<br><a href="#">firefox2.png</a>                                | Rendered correctly                 |
| Safari (iOS)            | iPhone 17 / iOS 26.4; iPhone 16 / iOS 27.0 | Login;<br>Order<br>History | <a href="#">safari1.png</a> ;<br><a href="#">safari2.png</a> ;<br><a href="#">safari3.png</a> | Rendered correctly (mobile reflow) |

The Safari coverage is a genuine Safari-on-iOS run, satisfying the strict "Chrome, Firefox, and Safari" requirement without a WebKit-on-Windows substitute. The flow rendered consistently on all three platforms; no platform-specific layout break, overflow, or missing control was found. The US-style date ( [7/31/2026](#) ) and comma- grouped currency ( [62,000,000 đ](#) ) appear identically on every platform, confirming they are the application-level locale defects already logged in Task 1. Full detail and the comparison are in [task-3-cross-platform/test-summary.md](#) .

## 10. Limitations

---

The GUI checklist (design, execution, and bugs), the seven usability sessions with analysis, and the three-platform cross-browser evidence are complete. Remaining before final submission: PDF exports of the reports, the Agent-Skill demonstration videos, and the self-assessed grade. The two Blocked checklist items still require a real screen reader, and all sessions ran on laptop/desktop, so mobile responsiveness was not evaluated on a real handset.